

# KI-Agenten Trainingsanalyse - Modellvergleich (final)

Stand: 2026-07-11 11-01 | Laufzeit: 44h 22m | Run: model\_comparison\_final\_v2

| Metrik                 | LSTM       | Transformer | MLP        |
|------------------------|------------|-------------|------------|
| Steps                  | 14,570,000 | 14,580,000  | 14,580,000 |
| Fortschritt            | 48.57%     | 48.60%      | 48.60%     |
| Reward (aktuell)       | 20.386     | 3.107       | 5.754      |
| Curriculum Phase       | 7          | 7           | 7          |
| Episode in Phase       | 7,127      | 7,339       | 7,460      |
| Gate-SuccessRate       | 45.4%      | 44.4%       | 44.5%      |
| Rolling Success        | 70.8%      | 28.8%       | 38.6%      |
| Beste Success          | 94.2%      | 54.1%       | 91.8%      |
| Lava-Crossings (Summe) | 0.000      | 0.066       | 0.000      |
| Lava-Sprungversuche    | 0.521      | 0.732       | 0.672      |
| Sprünge/Episode        | 10.658     | 15.127      | 18.787     |
| Tod: Lava              | 6.8%       | 36.6%       | 26.2%      |
| Tod: Timeout           | 2.7%       | 9.9%        | 13.1%      |

# Reward & Erfolgsrate

## Kumulativer Reward



## Rolling Success Rate (%)



# Curriculum & Todesursachen

## Curriculum Phase



## Gate-SuccessRate [%] (geteilt)



## Tod durch Timeout [%]



## Tod durch Lava [%]

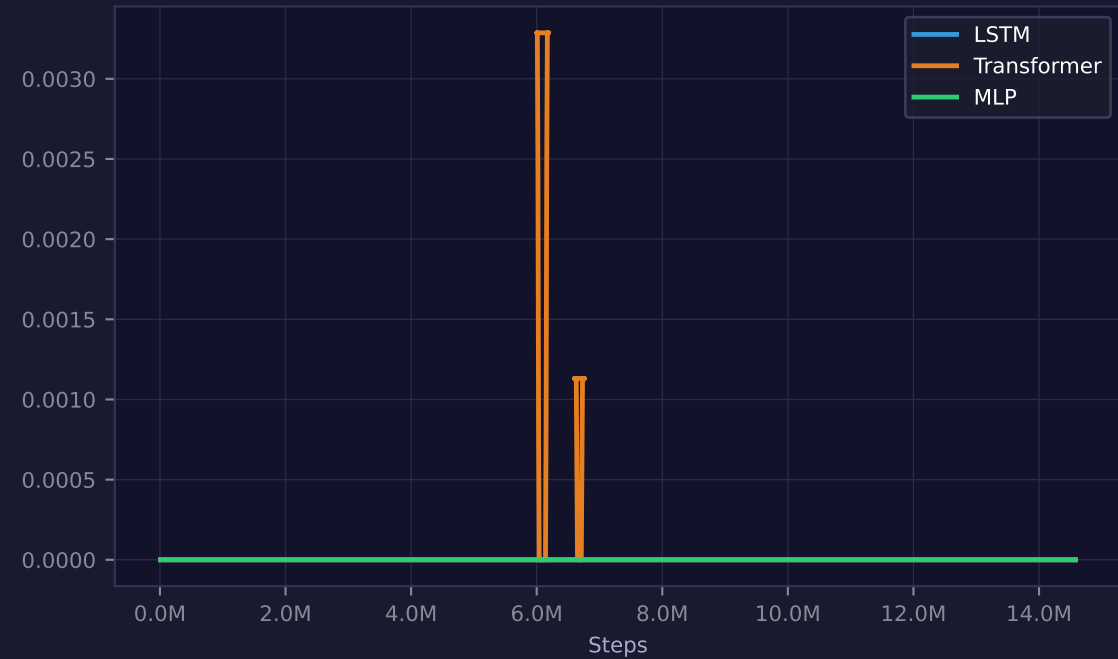


# Lava-Fokus: Lernt der Agent den Sprung?

Lava-Sprungversuche / Episode



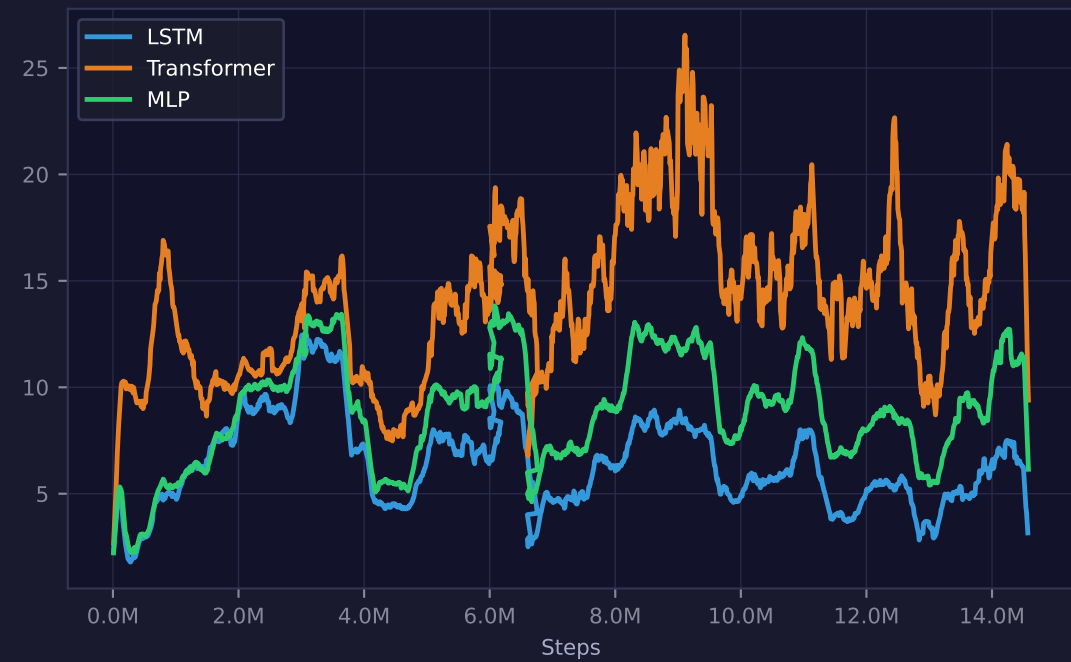
Erfolgreiche Überquerungen / Episode



Sprünge gesamt / Episode



Distanz zum Ziel bei Episodenende



# Reward-Zerlegung: Woher kommt der Reward?

PBRS-Shaping-Summe / Episode



Line-of-Sight-Reward-Summe / Episode



Curiosity Reward



Distanz-Fortschritt [%] (Start→Ende)

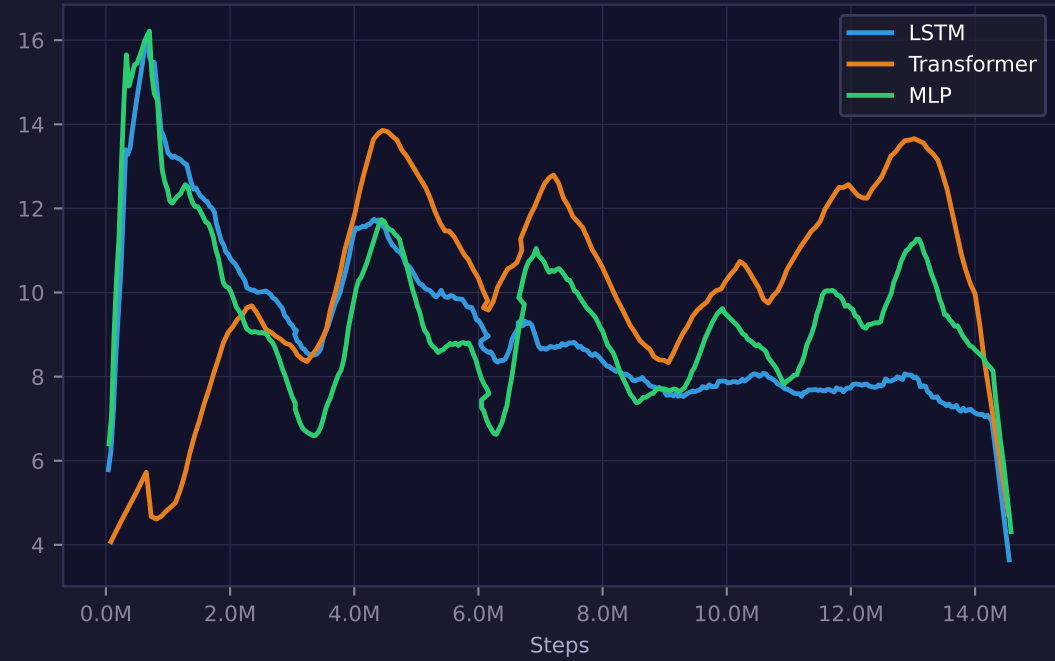


# Policy-Metriken

## Policy Loss



## Value Loss



## Entropy



## Learning Rate

